

In E-R diagram, a weak entity set is represented by ().

选择一项：

- ☐ A. rectangle
- ☐ B. diamond
- ☒ C. double rectangle ✓
- ☐ D. double diamond

Given a table for students selection of courses: SC (s_ID, c_ID, Grade). Please complete the following SQL statement that should find the average grade of the students who select the course with ID equal to 1.

```
SELECT [ ]  
FROM SC  
WHERE [ ]
```

Please fill the answers for the two blanks in [], separated by a blank space. Please do not include any blank space at either side of "=".

答案：



Given the following relations: Student (S#, SNAME, SEX, AGE), Course (C#, CNAME, TEACHER), SC (S#, C#, GRADE). To find the names of all female students who select the 'Computer' course, will involve the following relations: ()

选择一项：

- ☒ A. Student, Course, SC ✓
- ☐ B. Student, Course
- ☐ C. Student, SC
- ☐ D. Student

When we need to insert data to a table, if the data to be inserted are from other tables, we should use ().

选择一项：

- ☐ A. INSERT...VALUES
- ☐ B. all the other three are ok
- ☒ C. INSERT...SELECT ✓
- ☐ D. INSERT...DEFAULT

The three core elements of the ER model are: ()

选择一项:

- ☐ A. entities, relationships, keys
- ☒ B. entities, relationships, attributes ✓
- ☐ C. entities, entity sets, attributes
- ☐ D. entities, keys, domain

Trigger is a procedure that enables more complicated integrity constraints than the Check constraint.

选择一项:

- ☒ 对 ✓
- ☐ 错

Given a table Student(s_ID, Sname, Sage). Please complete the following SQL statement that should update the age of the student (whose ID is 201705001) to 22.

UPDATE Student

[]

WHERE Sno='201705001';

Please do not include any blank space at either side of "=".

答案: SET Sage=22



Consider the Cartesian-Product of two relations: $r1 \times r2$. Suppose the schema of $r1$ and $r2$ consists of 3 and 5 attributes respectively, and the number of tuples for them is 10 and 20 respectively.

Then the number of attributes of $r1 \times r2$ would be [], and the number of tuples of $r1 \times r2$ would be [].

Please fill in the answers for the two blanks in [], separated by a blank space.

答案: 8 200



A Database management system provides the functionality of querying, inserting, updating and deleting data in the database. This functionality is called ().

选择一项:

- ☒ A. Data manipulation ✓
- ☐ B. Data control
- ☐ C. Data protection
- ☐ D. Data definition

Given a table SC(s_ID, c_ID, Grade) for students selecting courses. Please complete the following SQL statement, which should be able to find from SC the ID of students who select only one course.

```
select s_ID
from SC
group by [ ]
having [ ]
```

Notes: Please fill in your answers for the above two blanks in [], separated by a blank space. Please do not include any blank space at either side of "=".

答案: s_ID COUNT(c_ID)=1 ✓

In relational algebra, which of the following operations extracts from a relation all tuples satisfying a specific condition? ()

选择一项:

- ☐ A. project
- ☒ B. select ✓
- ☐ C. join
- ☐ D. Cartesian-Product

In relational algebra, which of the following relational operations works on columns of a relation? ()

选择一项:

- ☐ A. union
- ☒ B. project ✓
- ☐ C. difference
- ☐ D. select

Given a table for students selection of courses: SC(s_ID, c_ID, Grade), a table for student information: Student(S_ID, Sname, Sage, Ssex), and a table for courses: Course(c_ID, Cname). Please complete the following SQL statement that should find all students' student ID, name, courses selected and grades.

```
select Student.s_ID, Sname, Cname, Grade
from Student, SC, [ ]
```

```
where Student.s_ID=SC.s_ID and [ ]
```

Please fill in the answers for the above two blanks, separated by a blank space. Do not add any blank space to either side of “=”.

答案: Course SC.c_ID=Course.c_ID



In the process of converting the E-R diagram to relational model, an one-to-many relationship set should be turned into an independent relation schema.

选择一项:

- ☒ 对 
- ☐ 错

When creating a table SC(student_ID, Course_ID, grade), which of the following integrity constraint should be used to meet the requirement that the students in SC must be those already in the Students table. ()

选择一项:

- ☐ A. default
- ☒ B. foreign key constraint 
- ☐ C. check constraint
- ☐ D. uniqueness constraint

To convert a many-to-many relationship set in the ER model to a relation schema, the primary key of the relation scheme should be ().

选择一项:

- ☐ A. attributes of two participating entity sets
- ☐ B. the discriminator attributes of the relationship set
- ☒ C. the primary keys of two participating entity sets and the discriminator attributes of the relationship set 
- ☐ D. the primary key of any one of the participating entity sets

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Given a table for the courses that the instructors teach: Teach(teacher_ID, course_ID, credit).

Create an INSTEAD OF INSERT trigger to ensure that, when inserting new tuples, each instructor teaches no more than 3 courses, otherwise create a warning message and refuse the insertion.

```
CREATE TRIGGER Morethanthree ON Teach
```

```
INSTEAD OF INSERT
```

```
AS
```

```
BEGIN
```

```
    IF [    ] (
```

```
        SELECT inserted.teacher_ID
```

```
        FROM inserted INNER JOIN Teach
```

```
            ON inserted.teacher_ID=Teach.teacher_ID
```

```
        GROUP BY inserted.teacher_ID
```

```
        HAVING COUNT(Teach.Course_ID) + COUNT(inserted.Course_ID) > 3 )
```

```
BEGIN
```

```
    RAISERROR('The maximum number of courses an instructor teaches is 3!', 16, 1);
```

```
    RETURN;
```

```
END
```

```
INSERT INTO Teach (teacher_ID, Course_ID, credit)
```

```
SELECT teacher_ID, Course_ID, credit
```

```
FROM [    ];
```

```
END;
```

```
GO
```

** Please fill in the above two blanks in [], separated by a blank space.

答案:

Suppose we have created a table Course using the following SQL statement: Create table Course (CID char(10) primary key, CN char(20) not null, CC int not null). The following tuple can be inserted into the table Course. ()

选择一项:

- ☐ A. (005, 'VC++', NULL)
- ☒ B. ('005', 'VC++', 6) ✓
- ☐ C. ('005', NULL, 6)
- ☐ D. (NULL, 'VC++', 6)

In relational model, the primary key of a relation ()

选择一项:

- ☐ A. is an arbitrary set of attributes
- ☐ B. contains at most one attribute
- ☐ C. is any set of attributes whose values are sufficient to uniquely identify any tuple in the relation
- ☒ D. is a minimum set of attributes chosen by the database designer, and their values can uniquely identify any tuple in the relation ✓

The integrity constraints that can be specified when creating a table in SQL do not include ().

选择一项:

- ☒ A. triggers ✓
- ☐ B. referential integrity constraints
- ☐ C. user defined constraints
- ☐ D. primary key constraints

21. Given two tables: dept(deptno, dname), emp(empno, ename, phone, deptno), where deptno in emp is a foreign key referencing the dept table. Which of the following SQL statements is correct to find the empno and ename information of all the employees in the department (dname) 'finance'? ()

选择一项:

A.

SELECT empno,ename FROM emp

WHERE deptno=(SELECT * FROM dept WHERE dname='finance')

B.

SELECT empno,ename FROM emp

WHERE empno=(SELECT empno FROM dept WHERE dname='finance')

C.

SELECT empno,ename FROM dept

WHERE deptno=(SELECT deptno FROM emp WHERE dname='finance')

D.

SELECT empno,ename FROM emp

WHERE deptno=(SELECT deptno FROM dept WHERE dname='finance')

Given the following relations: Patient (P_ID, P_name, Gender, DoB) with primary key P_ID; and Diagnosis (P_ID, Doctor_ID, Doctor_name, Date, Result) with primary key Doctor_ID. The foreign key of the Diagnosis relation is ().

选择一项：

- ☐ A. Doctor_name
- ☐ B. P_ID, P_name
- ☐ C. Doctor_ID
- ☒ D. P_ID ✓

Given a product table Product (pID, pName, price). Suppose we need to find the information about those products whose prices are greater than all the products named (pName) 'noodle'. ()

选择一项：

- ☐ A. SELECT * FROM product WHERE EXISTS pName='noodle'
- ☐ B. SELECT * FROM product WHERE price>(SELECT max(*) FROM product WHERE pName='noodle')
- ☒ C. SELECT * FROM product WHERE price>(SELECT max(price) FROM product WHERE pName='noodle') ✓
- ☐ D. SELECT * FROM product WHERE max(price)>'noodle'

In the process of relational database design, the design of relation scheme is a task of the () phase.

选择一项：

- ☐ A. conceptual design
- ☒ B. logical design ✓
- ☐ C. physical design
- ☐ D. requirement analysis

In a database system, the meaning of the NULL value is ().

选择一项：

- ☐ A. 0
- ☐ B. any value
- ☐ C. a string of length 0
- ☒ D. unknown value ✓

Assume each student can select multiple courses, and a course can be selected by multiple students. Then there exist () relationships between the entity sets student and course.

选择一项：

- ☐ A. one to many
- ☐ B. many to one
- ☐ C. one to one
- ☒ D. many to many ✓

Which of the following statements about temporary tables DELETED and INSERTED used in trigger is incorrect ().

选择一项：

- ☐ A. The DELETED table does not share any tuple with the table on which the trigger is created.
- ☐ B. The UPDATE operation needs both DELETED and INSERTED tables.
- ☒ C. The INSERTED table has the same tuples as the table on which the trigger is created. ✓
- ☐ D. DELETED and INSERTED are of the same scheme as the table on which the trigger is created.